

Test Case Table – Patch/Vulnerability Report Builder:

Test Case ID	Test Case Name	Input	Expected Result	Status
TC001	JSON File Upload	Valid vulnerability_scan.json	File uploaded successfully	Pass
TC002	CSV File Upload	Valid vulnerability_scan.csv	File uploaded successfully	Pass
TC003	CVE Extraction	Input file containing CVE IDs	All CVE IDs extracted correctly	Pass
TC004	NVD API Integration	CVE-2024-3094	Vulnerability details fetched from NVD API	Pass
TC005	Risk Ranking	Multiple CVEs with different CVSS scores	Vulnerabilities ranked from highest to lowest risk	Pass
TC006	Gemini AI Analysis	Enriched vulnerability data	Business impact and recommendations generated	Pass
TC007	Executive Summary Generation	Processed vulnerability dataset	Executive summary generated successfully	Pass
TC008	Dashboard Generation	Analyzed vulnerability data	Charts and statistics displayed correctly	Pass
TC009	Markdown Report Download	Completed analysis	Markdown report generated and downloadable	Pass
TC010	PDF Report Download	Completed analysis	PDF report generated and downloadable	Pass
TC011	Top-20 Risk Identification	Dataset with multiple vulnerabilities	Top 20 highest-risk vulnerabilities displayed	Pass
TC012	Pytest Execution	Run pytest command	All test cases executed successfully	Pass

Mapping to Your Project Files:

Test Function	Purpose
test_json_parser()	Validate JSON input parsing
test_csv_parser()	Validate CSV input parsing
test_nvd_api()	Verify NVD API integration
test_risk_ranking()	Verify vulnerability ranking logic
test_report_generation()	Verify report creation
test_ai_summary()	Verify AI-generated summaries

Short Version:

Test Case	Result
JSON Parsing	Pass
CSV Parsing	Pass
NVD API Fetch	Pass
Risk Ranking	Pass
AI Summary Generation	Pass
Report Generation	Pass
PDF Export	Pass
Dashboard Visualization	Pass